LAB EXPERIMENT

5.Write a Python function to analyze the histogram of the given input image based on color levels using Open CV.

Code:

```
import cv2
import matplotlib.pyplot as plt

# Read the input image
img = cv2.imread(r"C:\Users\Admin\Downloads\images.jpg")

# Check whether the image is loaded
if img is None:
    print("Error: Image not found!")
else:
    # Display the original image
    cv2.imshow("Original Image", img)

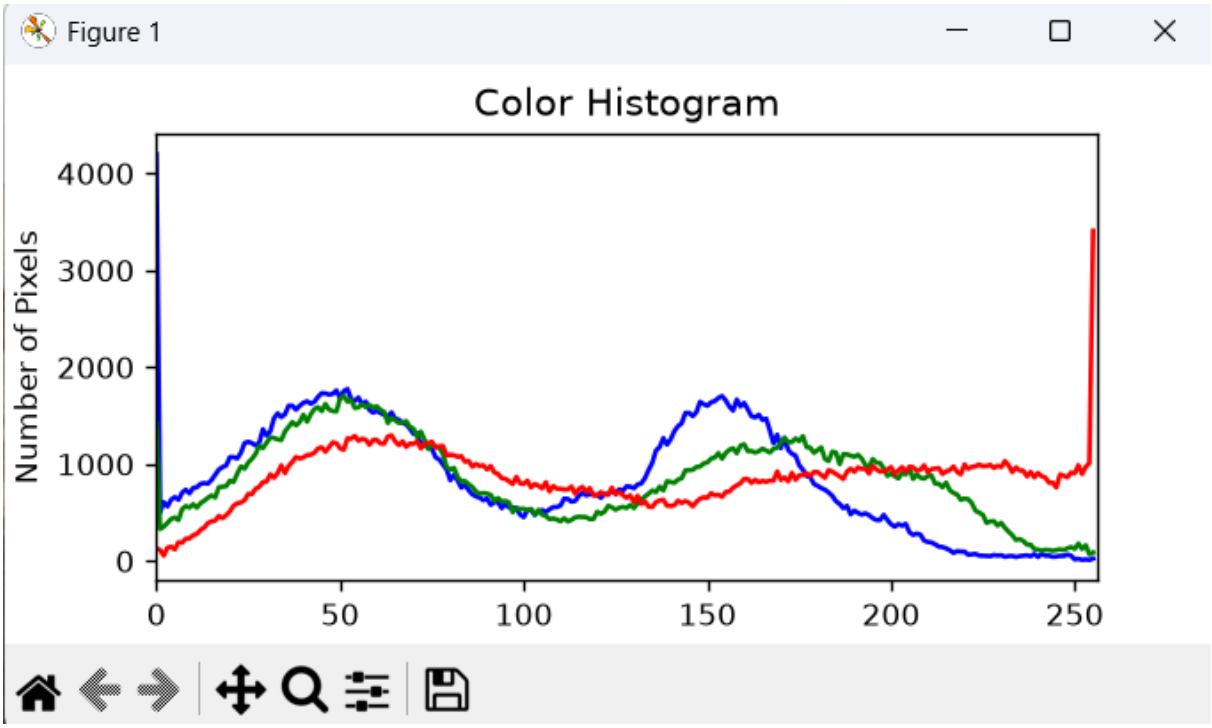
    # Calculate and display histogram for each color channel
    colors = ('b', 'g', 'r')

    for i, color in enumerate(colors):
        hist = cv2.calcHist([img], [i], None, [256], [0, 256])
        plt.plot(hist, color=color)

    plt.title("Color Histogram")
    plt.xlabel("Pixel Intensity (0-255)")
    plt.ylabel("Number of Pixels")
    plt.xlim([0, 256])
    plt.show()

    cv2.waitKey(0)
    cv2.destroyAllWindows()
```


Original Image


Figure 1
Color Histogram
Number of Pixels
4000
3000
2000
1000
0
0
50
100
150
200
250